

CHAPTER 13

Product Quality Spread Trading

Trading the price gaps between grades — strategies built on product specifications and differences in quality.



SAME NAME, DIFFERENT VALUE

"Fuel oil" is not one price — quality is

Two cargoes can carry the same product name yet trade dollars apart, because they meet different specifications — sulfur, density, octane, cold-flow. The market prices that difference, and the gap between two qualities is itself a tradable spread.

\$0 → \$315/t

the low- vs high-sulfur fuel-oil spread when IMO 2020 hit

A spec is a price

Lower sulfur, higher octane or lighter density usually commands a premium — the spread is the market valuing quality.

A trade in its own right

You can buy the cheaper grade and sell the dearer one, betting the quality gap widens or narrows.

THE DEFINITION

What a quality spread is

It is the price difference between two grades of the same broad product that differ only in specification — the market's price tag on quality itself.

High-quality grade

low sulfur · high octane · light
price \$\$\$

QUALITY SPREAD

= high grade – low grade

Low-quality grade

high sulfur · low octane · heavy
price \$

It isolates one variable — quality. Strip out location and time, and what's left is purely the market's reward for a better specification.

WHAT "QUALITY" MEANS

The dimensions of quality

A handful of measurable properties separate a premium grade from a discounted one. Each is a specification a buyer pays up — or down — for.



Sulfur

Low-sulfur grades are cleaner and meet tighter fuel rules, so they trade at a premium to high-sulfur ones.



Density (light / heavy)

Lighter grades yield more high-value product with less processing, commanding more than heavy ones.



Octane / cetane

Higher octane in gasoline (and cetane in diesel) means better performance — and a price premium.



Cold-flow & volatility

Pour point, cloud point and vapour pressure decide where and when a grade can be used — and what it's worth there.

THE MARQUEE EXAMPLE · MARINE FUEL

The sulfur spread — VLSFO vs HSFO

HI-5 SPREAD (VLSFO – HSFO), \$/TONNE



Created by regulation

From Jan 2020, IMO capped ships' fuel sulfur at 0.5%. Overnight, clean VLSFO commanded a huge premium over dirty HSFO.

Violently variable

The spread spiked to ~\$315/t, collapsed to \$45 in Covid, hit \$420 in 2022 — a quality gap that trades like a market of its own.

Why traders watch it

It drives whether shipowners fit scrubbers and burn cheap HSFO, or pay up for VLSFO — and refiners hedge it directly.

BEYOND SULFUR

Two more quality spreads

The same idea repeats across the barrel — any measurable quality difference between two grades is a spread someone trades.

DENSITY

Light – heavy crude

e.g. Brent – Dubai · WTI – WCS

Light crude yields more high-value product, so it trades above heavy crude. The gap widens when light products are in demand and narrows when heavy crude is scarce.

OCTANE

Premium – regular gasoline

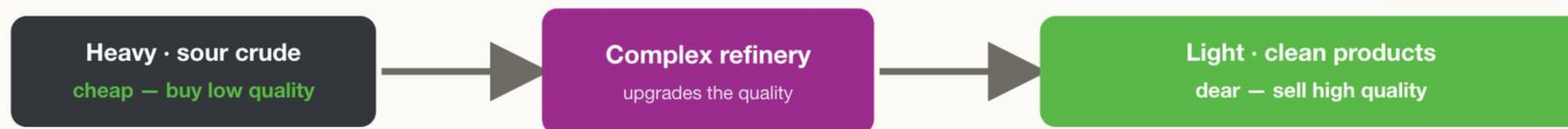
the octane spread

Higher-octane premium gasoline sells above regular. The spread reflects how much the market will pay for performance — and swings with refinery octane supply and demand.

WHO'S ON THE OTHER SIDE

The refiner's edge in quality

Quality spreads are not abstract — a complex refinery exists precisely to capture them, buying low quality and selling high.



The quality uplift is the refiner's margin. When a quality spread widens, complex refiners that can do this upgrade earn more — which is exactly why they hedge it.

PUTTING ON THE TRADE

How quality spreads are traded

Whether to hedge a real exposure or to speculate, the mechanics are the same: take opposite positions in the two qualities.

Buy one grade, sell the other

Go long the grade you expect to outperform and short the other — your P&L is the change in the spread, not the flat price.

Refiners hedge the differential

A refiner upgrading low into high quality locks in the spread it earns, protecting its margin if quality premiums shrink.

Paper markers & swaps

Many quality spreads (Hi-5, light-heavy, octane) trade as listed differentials or OTC swaps — no cargo required.

A cleaner bet

Because both legs move with the wider market, flat-price risk cancels — leaving a focused view on quality alone.

DRIVERS

What moves quality spreads

Quality premiums are not fixed — they breathe with rules, refining and demand.



Regulation

A new sulfur cap or fuel standard can create or destroy a premium overnight — IMO 2020 is the textbook case.



Refining capacity

When upgrading units run flat out or go down for maintenance, the supply of high-quality grades tightens and spreads widen.



Crude slate

The quality of crude being run shapes how much premium product comes out — and what's left over as discount material.



Season & demand

Summer octane demand or a cold-weather diesel pull can lift one grade's premium independently of the rest.

KNOW THE DOWNSIDE

The risks of quality-spread trading

A clean idea on paper still carries real hazards once the two grades stop moving together.

Correlation can break

Most of the time the two grades track each other — until a shock decouples them and the spread lurches against you.

Liquidity

Some quality differentials trade thinly, so getting in or out can be costly and slow when you most need to.

Basis & blending

A traded marker may not exactly match the physical grade you hold — leaving a residual gap that doesn't hedge.

Regulatory whiplash

The same rules that create a premium can change again, repricing a spread faster than positions can adjust.

CONCLUSION — TRADING THE SPEC

Quality is a price you can isolate and trade.

A spec is a premium

Sulfur, density and octane each carry a price — the gap between two grades is the quality spread.

Rules reset the gap

Regulation like IMO 2020 can create or erase a quality premium almost overnight.

It's the refiner's margin

Upgrading cheap low quality into dear high quality is exactly what the quality spread pays for.